A Pathway zonation in bovine liver
(dark = called, light = weak, grey = none/mixed; ✓/✗ vs rodent)

Lactate / glycerol entry [rodent PP]
Urea cycle [rodent PP]
Gluconeogenesis core [rodent PP]
Succinyl-CoA ligase / TCA entry [rodent PP]
OXPHOS (complex I/III/IV/V) [rodent PP]
Glucogenic amino acid catabolism [rodent PP]
Propionyl-CoA -> succinyl-CoA [rodent unknown]
Mitochondrial beta-oxidation [rodent PP]
Ketogenesis [rodent PP]
SCFA activation (acyl-CoA synthetases) [rodent mixed]
VLDL assembly / export [rodent mixed]
Wnt / pericentral identity [rodent PC]
Iron / hepcidin [rodent PP]
Glycolysis [rodent PC]
De novo lipogenesis [rodent PC]
Bile acid synthesis [rodent PC]
Xenobiotic CYPs [rodent PC]
Glutamine synthesis / NH3 scavenging [rodent PC]

gene-mean log2FC
expression-matched null

−1.5 −1.0 −0.5 0.0 0.5
log2 FC portal / central (bars: pooled counts, block-adjusted)

B Pooled pathway profiles
(dotted = weak / mixed / not zonated)

Propionyl-CoA -> succinyl-CoA [not zonated]
Gluconeogenesis core [PP]
Urea cycle [PP]
Glutamine synthesis / NH3 scavenging [PC]
Mitochondrial beta-oxidation [not zonated]
Ketogenesis [not zonated]
De novo lipogenesis [mixed]
Bile acid synthesis [PC]
OXPHOS (complex I/III/IV/V) [not zonated]

1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75
log2 relative expression (block-adjusted)
0.2 0.4 0.6 0.8
η (0 = central vein, 1 = portal tract)

C Propionate → glucose and
N-handling genes (* FDR<0.05)

ACSM1
ACSS3 *
ACSM2B
ACSS2
ACSM3
ACSM5
MCEE
PCCB
MMAB
PCCA
MMUT
SUCLG2
SUCLG1
SLC37A4
PCK2
PC
FBP1 *
G6PC1
PCK1 *
NAGS
CPS1
ASL *
OTC
ARG1 *
ASS1 *
RHBG *
GLUL *
SLC1A2
HMGCL
HMGCS2
ACAT1

SCFA activation (acyl-CoA synthetases)
Propionyl-CoA -> succinyl-CoA
Succinyl-CoA ligase / TCA entry
Gluconeogenesis core
Urea cycle
Glutamine synthesis / NH3 scavenging
Ketogenesis

0.8 0.6 0.4 0.2 0.0 −0.2 −0.4 −0.6 −0.8
log2 relative expression

0.06 0.19 0.31 0.44 0.56 0.69 0.81 0.94
η layer (central → portal)

D Every tested pathway gene (coloured = FDR<0.05; background = rodent expectation). Pericentral programmes keep rodent-like gradients; propionate entry, β-oxidation, ketogenesis are flat and CPS1 is not periportal

2 1 0 −1 −2
gene log2 FC portal / central

GLS2
HAMP
ACACA
HAL
ASS1
GPD1
PCK1
G6PC1
ARG1
ASL
FBP1
CPS1
HMGCS2
HPX
TF
PC
CPT1A
ACSS3
MMUT
PCCA
PCCB
CYP2C89
CYP1A2
CYP3A76
CYP8B1
CYP27A1
CYP7A1
SCD
ELOVL6
CYP2E1
GLUL
CYP2A13
TBX3
ZNRF3
RHBG

Glutamine synthesis / NH3 scavenging [rodent PC]
Xenobiotic CYPs [rodent PC]
Bile acid synthesis [rodent PC]
De novo lipogenesis [rodent PC]
Glycolysis [rodent PC]
Wnt / pericentral identity [rodent PC]
Iron / hepcidin [rodent PP]
Ketogenesis [rodent PP]
Mitochondrial beta-oxidation [rodent PP]
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